

# Electrical Fuse Schedule (Implementation - Lynx Topology)

As-of date: 2026-08-08

Purpose: define each required fuse by circuit, protected conductor/device, holder/housing method, physical placement, and linked wire-gauge assumptions for the approved Phase 1 Lynx architecture with a battery-backed 12V bus and dedicated 48V secondary alternator branch.

Related docs:

- Canonical electrical/system baseline: docs/core/SYSTEMS.md
- Implementation topology and conductor map: docs/implementation/ELECTRICAL\_overview\_diagram.md
- Decisions/open items tracker: docs/core/TRACKING.md
- BOM source of truth: bom/bom\_estimated\_items.csv

## Design Basis

- Topology: Victron Lynx Distributor M10 ( LYN060102010 ) with 4 active fused 48V branches; Slot 4 feeds Orion input.
- Battery bank assumption: 3x 48V 100Ah batteries in parallel ( 3 separate battery-positive conductors leaving batteries).
- 12V distribution assumption: 12V fuse block used as the shared junction device (main + stud = source combine, integrated negative bus/main - = return), fed by Orion-Tr Smart 48/12-30 charger and a 12V 100Ah LiFeP04 buffer battery branch.
- Parallel-bank safety rule: use one Class T fuse per battery-positive conductor leaving the battery.
- Active branch devices on Lynx:

1. MultiPlus-II 48/3000
2. SmartSolar 150/45
3. Dedicated 48V secondary alternator branch (Mechman/WS500 path)
4. Slot 4 / F-05 — Orion-Tr 48/12-30 input through one 40A MEGA body-marked at least 58VDC ; standalone F-06 is retired.

- Current shop identification note: do not infer the purpose of the physically X-marked/misrated Lynx fuse from the X alone. Identify its slot. Slot 2 requires F-03 60A/80V MEGA for the MPPT; Slot 4 requires F-05 40A MEGA marked >=58VDC for Orion. No 32V -rated fuse is acceptable on an energized 48V branch.
- Alternator path lock for this pass:

1. F-04 is locked to 200A/80V MEGA in Lynx Slot 3 at the house-bank/Lynx end of the alternator positive run. Mechman's published 48V Elite curve reaches about 145.7A ; the 125% fuse basis is about 182A , so 200A is the next standard size and remains within the installed 2/0 AWG conductor envelope.
2. Obsolete pre-Mechman charger/fuse paths are removed from active architecture and primary layout planning.
3. Confirmed PH-VAN harness uses one short red/black pair as combined regulator power and voltage sense. Treat former separate F-12 regulator-power and F-13 positive-sense functions as one 15A Class-CC fused red lead from the F-04 alternator/load-side stud. Put the Wakespeed shunt in the common battery-negative path after the SmartShunt; its purple/high and grey/low sense leads remain unfused.
4. Ford Upfitter Switch #3 is locked as the manual WS500 enable source through F-15 .

## Alternator Branch Fuse + Wire Finalization ( 2026-03-19 )

Assumptions for this pass:

1. One-way alternator run length: 20 ft .
2. Dedicated positive and dedicated negative runs (loop length 40 ft ).
3. Charging target voltage basis: 58.4V .
4. Voltage-drop planning target: about <=2% under expected charging current.

Resistance and drop screen (  $V_{drop} = I * (2 * L * R_{per\_ft})$  ):

| Gauge   | 150A drop | % @ 58.4V | 200A drop | % @ 58.4V | Decision note                            |
|---------|-----------|-----------|-----------|-----------|------------------------------------------|
| 4 AWG   | 1.49V     | 2.55%     | 1.99V     | 3.40%     | Reject for this run length/current class |
| 2 AWG   | 0.94V     | 1.61%     | 1.25V     | 2.14%     | Acceptable for ~150A class               |
| 1/0 AWG | 0.59V     | 1.01%     | 0.79V     | 1.35%     | Strong margin                            |
| 2/0 AWG | 0.47V     | 0.80%     | 0.62V     | 1.07%     | Best margin; inventory already on hand   |

Lock for this build pass:

- Use F-04 200A/80V in Lynx Slot 3. This preserves margin above the published 145.7A curve while protecting the existing 2/0 AWG run.

- Reuse existing uncut 2/0 inventory for alternator + and dedicated negative run.
- Inactive BOM row 173 preserves the removed contingency estimate; open a new active line only if field measurement proves an actual shortfall.

## Required Fuse Map (Start-to-Finish, With Housing)

| Fuse ID | Circuit (source -> load)                              | Protected wire/device                     | Fuse type and voltage class                                                      | Amperage                                                                                                                   | Holder or housing method                                           | Physical location                                             | Planned conductor gauge                            |
|---------|-------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|
| F-01A   | Battery A + -> bank positive combine/disconnect input | Battery A positive cable leaving battery  | Class T (>=125VDC)                                                               | 200A provisional                                                                                                           | Blue Sea Class T fuse block (covered stud mount, 110A-200A family) | Battery compartment, within ~`7"`` of Battery A positive post | 2/0 AWG                                            |
| F-01B   | Battery B + -> bank positive combine/disconnect input | Battery B positive cable leaving battery  | Class T (>=125VDC)                                                               | 200A provisional                                                                                                           | Blue Sea Class T fuse block (covered stud mount, 110A-200A family) | Battery compartment, within ~`7"`` of Battery B positive post | 2/0 AWG                                            |
| F-01C   | Battery C + -> bank positive combine/disconnect input | Battery C positive cable leaving battery  | Class T (>=125VDC)                                                               | 200A provisional                                                                                                           | Blue Sea Class T fuse block (covered stud mount, 110A-200A family) | Battery compartment, within ~`7"`` of Battery C positive post | 2/0 AWG                                            |
| F-02    | Lynx Slot 1 -> MultiPlus DC+                          | Main inverter positive feeder             | MEGA, 58V or 80V                                                                 | 125A                                                                                                                       | Integrated Lynx Distributor fuse slot                              | Lynx Distributor, Slot 1                                      | 2/0 AWG planned (AWG 1 minimum on short run)       |
| F-03    | Lynx Slot 2 -> SmartSolar BAT+                        | MPPT battery-side positive feeder         | MEGA, 80V Victron replacement stock                                              | 60A                                                                                                                        | Integrated Lynx Distributor fuse slot                              | Lynx Distributor, Slot 2                                      | 6 AWG                                              |
| F-04    | Lynx Slot 3 -> 48V alternator branch input (ALT+)     | Alternator-to-house positive charge cable | MEGA, 80V                                                                        | 200A                                                                                                                       | Integrated Lynx Distributor fuse slot                              | Lynx Distributor, Slot 3                                      | 2/0 AWG (reuse lock)                               |
| F-05    | Lynx Slot 4 -> Orion 48V input +                      | Orion 48V input feeder                    | MEGA 40A, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback | 40A deliberate feeder-protection value; above Victron's 20A table recommendation but safely below installed 6 AWG ampacity | Integrated Lynx Distributor fuse slot                              | Lynx Distributor, Slot 4                                      | Existing AWG direct to Orion; no second input fuse |

|                   |                                                                           |                                                                                      |                                                                                                       |                        |                                                                                                                                                     |                                                          |                                                                                                           |
|-------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| F-06              | Retired standalone Orion input fuse position                              | Not active                                                                           | N/A                                                                                                   | Not installed          | No holder                                                                                                                                           | N/A                                                      | Purchased 30A/58V MIDI and FKS/ATO stock planned; propose DIN hardware remain unused; not stock with F-07 |
| F-07              | Orion 12V output + - > 12V fuse block main + stud                         | Main 12V feeder from Orion into shared source-combine point                          | MEGA, 80V Victron replacement stock                                                                   | 60A                    | Victron MEGA fuse holder (external, non-Lynx)                                                                                                       | Electrical cabinet, within ~`7" of Orion12V` output stud | 6 AWG planned; 4 AWG minimum per Orion cable table)                                                       |
| F-09A/B/C         | PV string + leads -> MPPT PV combiner                                     | Each solar string positive conductor and reverse-current path                        | gPV string fuse (>=150VDC)                                                                            | 15A each (provisional) | 10x38 touch-safe PV fuse holders in weatherproof combiner enclosure                                                                                 | Roof-entry combiner near gland/pass-through              | 10 AWG PV wire                                                                                            |
| F-10              | 12V fuse block branch circuits -> each 12V load                           | Individual 12V branch conductors and load circuits                                   | ATO/ATC blade fuses (32V class)                                                                       | Per-circuit            | Integrated sockets in marine 12V fuse block                                                                                                         | 12V fuse block in electrical cabinet                     | Per branch                                                                                                |
| AUDIO-HU / 12V-12 | 12V fuse block -> Kicker 46KMC2 media center/source unit                  | KMC2 source/head-unit branch conductor and device harness                            | ATO/ATC branch fuse (32V class) plus KMC2 harness 15A ATM fuse                                        | 15A                    | Integrated 12V fuse block branch plus OEM KMC2 harness fuse                                                                                         | Electrical cabinet to driver-side DC shelf/source face   | 12 AWG duplex if kept around 10 ft; 10 AWG route grows toward 12 ft+                                      |
| AUDIO-SUB         | 12V source/main + stud -> external fuse -> Kicker 49PTRTP10 powered sub + | Powered-sub positive branch and subwoofer input; manual-required external protection | Inline/MRBF/AFS/ANL-class fuse matched to selected 4 AWG kit (32V+ DC class acceptable on 12V branch) | 40A                    | Holder within about 18 in of 12V source takeoff; if using Kicker 47KMPK4, fit the PTRTP10-required 40A fuse rather than the kit generic larger fuse | Near 12V source/junction, downstream of SW-12V-BATT      | 4 AWG positive with matching 4 AWG return to 12V negative bus/main stud                                   |
| F-11              | 12V buffer battery + -> 12V fuse block                                    | Buffer battery source cable and                                                      | Inline MIDI/AMI/ANL family rated >=32VDC                                                              | 100A class baseline    | Sealed inline holder mounted                                                                                                                        | Within ~`7" of 12V buffer                                | 4 AWG planned                                                                                             |

|                 |                                                                                 |                                                                |                                                                                                   |           |                                                                               |                                                                                                                                                                                          |                                 |
|-----------------|---------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|                 | main + stud via SW-12V-BATT                                                     | downstream junction fault exposure                             |                                                                                                   |           | close to battery positive                                                     | battery positive post                                                                                                                                                                    |                                 |
| F-12/F-13-PHVAN | F-04 alternator/load-side stud -> WS500 PH-VAN red lead                         | Combined WS500 regulator power and positive voltage-sense lead | Mersen ATDR15 time-delay Class CC, 300VDC                                                         | 15A       | Mersen USCC1 touch-safe 600VDC holder in a separate small DC DIN enclosure    | Electrical panel immediately beside F-04; short 14 AWG pigtails with sealed step-down splice to the 16 AWG harness                                                                       | 14 AWG pigtail / harness lead   |
| CERBO-PWR       | 48V system positive -> Cerbo GX power +                                         | Cerbo GX low-current electronics feed                          | Inline fuse/holder rated for 48V bank maximum voltage                                             | 1A-3A     | Small inline holder close to the positive takeoff                             | Electrical cabinet, preferably system/load side of main disconnect so Cerbo powers down with the house system during bench shutdown                                                      | 18 AWG red/black duplex accepta |
| F-15            | Ford Upfitter #3 -> WS500 brown ignition/enable wire                            | Low-current regulator enable/control wire                      | Inline ATC/ATO (32v class acceptable; 12V control circuit)                                        | 3A        | Sealed inline holder near the Ford upfitter blunt-cut source / splice handoff | Engine bay or control-wire handoff point before small-gauge run to WS500                                                                                                                 | 16 AWG TXL/GXL                  |
| OEM-SHUNT       | Battery-side positive or Lynx/system positive tap -> SmartShunt Vbatt+ terminal | SmartShunt electronics supply/sense lead                       | External Victron-supplied red cable with inline low-current fuse; not an internal SmartShunt fuse | OEM value | Inline holder in supplied red cable                                           | Prefer battery-side positive if SOC continuity is desired while the main disconnect is open; system side is acceptable if zero parasitic draw during disconnect-off storage matters more | OEM harness lead                |

## Retired Fuse IDs

Obsolete pre-Mechman charger/fuse paths are removed from the active schedule. Do not reserve board space or labels for them.

## Spare Fuse Inventory (Updated)

| Fuse type | Installed qty | Spare qty to carry | Notes |
|-----------|---------------|--------------------|-------|
|-----------|---------------|--------------------|-------|

|                                                                               |                                               |                                                         |                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Class T 200A (provisional installed)                                          | 3                                             | 1 on hand                                               | Owner confirmed 3 holders and 4 slow-blow Class T fuses total (3 installed + 1 spare); add more only if the one-spare-per-installed policy is later desired                                                              |
| MEGA 200A/80V                                                                 | 1                                             | 1 minimum                                               | Alternator branch (F-04); Amazon Victron stock is commonly sold as a five-pack, so install one and carry one while retaining the remainder as shop stock                                                                 |
| MEGA 125A (58V/80V)                                                           | 1                                             | 4                                                       | MultiPlus branch                                                                                                                                                                                                         |
| MEGA 60A (80V)                                                                | 2                                             | 3                                                       | Victron 5-pack row 188: install MPPT (F-03) + Orion output (F-07), keep 3 spares. Earlier 6x low-cost 60A MEGA batch was owner-confirmed misadvertised/not actually 58V and is quarantined/trash, not 48V install stock. |
| MEGA 40A, >=58VDC                                                             | 1 active                                      | 1-2                                                     | Orion input in Lynx Slot 4 (F-05); existing body-marked 58VDC stock is acceptable under the locked 56.8v charge ceiling; use Victron CIP138040020 40A/80V as replacement fallback                                        |
| Retired Orion standalone input-fuse stock                                     | 0 active                                      | Optional purchased stock only                           | 30A/58V MIDI and FKS/ATO parts are not installed; no USM1/ATM20 purchase required                                                                                                                                        |
| 12V buffer battery main fuse (100A class)                                     | 1                                             | 3                                                       | Spare pack basis is BOM row 105                                                                                                                                                                                          |
| WS500 PH-VAN combined regulator power / positive-sense fuse (F-12/F-13-PHVAN) | 1 required position                           | 1 optional later                                        | One Mersen ATDR15 15A/300VDC Class-CC fuse in a USCC1 holder is sufficient to commission; no proprietary automotive contacts are needed                                                                                  |
| Cerbo GX power fuse (CERBO-PWR)                                               | 1 active position                             | 1                                                       | Carry a spare 1A-3A fuse/holder rated for the 48v bank maximum                                                                                                                                                           |
| WS500 ignition/enable fuse (F-15)                                             | 1 active position                             | 2                                                       | Carry spare 3A mini/ATO fuse and one spare sealed holder; 12V control circuit                                                                                                                                            |
| PV string fuse 15A gPV                                                        | 3                                             | 3                                                       | One spare per string                                                                                                                                                                                                     |
| SmartShunt OEM harness fuse                                                   | 1                                             | 1                                                       | Keep OEM-equivalent spare if field-replaceable                                                                                                                                                                           |
| ATO/ATC branch fuses                                                          | variable                                      | 2 each used value                                       | Keep mixed kit onboard                                                                                                                                                                                                   |
| Camper audio KMC2 branch (AUDIO-HU)                                           | 1 active 15A branch plus KMC2 harness 15A ATM | 2 spare 15A blade/ATM fuses                             | Preserve both source-side conductor protection and the KMC2 harness fuse                                                                                                                                                 |
| Camper audio powered sub (AUDIO-SUB)                                          | 1 active 40A external fuse                    | 1-2 spare 40A fuses matching the selected holder family | PTRTP10 manual calls for 40A external fuse and 4 AWG power/ground                                                                                                                                                        |

## BOM Row Mapping

| Fuse scope                                                           | BOM row(s)                                             |
|----------------------------------------------------------------------|--------------------------------------------------------|
| Main battery Class T protection (F-01A/F-01B/F-01C) + Class T spares | bom/bom_estimated_items.csv row 7                      |
| Lynx branch MEGA fuses (F-02 to F-05 installed) + spare set          | bom/bom_estimated_items.csv rows 10, 170, 188, and 323 |
| Orion installed fuse-holder hardware (F-05 Lynx input, F-07 external | bom/bom_estimated_items.csv rows 6 and 11              |

|                                                                      |                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| output)                                                              |                                                                                                                                                                                                                                      |
| Retired Orion standalone input-fuse stock (F-06)                     | Purchased stock: active BOM rows 133, 182, and 321; inactive rows 326 and 230 preserve retired holder/bridge history. Row 321 empty FKS housings and row 133 20A fuses are not used anywhere in the final alternator/Orion topology. |
| WS500 regulator-power and voltage-sense protection (F-12/F-13-PHVAN) | Active BOM row 171; inactive row 320 preserves the retired separate-3A concept                                                                                                                                                       |
| Cerbo GX power feed (CERBO-PWR)                                      | bom/bom_estimated_items.csv row 22; small inline fuse/holder may come from low-current install stock                                                                                                                                 |
| WS500 Upfitter #3 enable/control path (F-15)                         | bom/bom_estimated_items.csv row 176                                                                                                                                                                                                  |
| 12V buffer battery (B12)                                             | bom/bom_estimated_items.csv row 21                                                                                                                                                                                                   |
| 12V buffer battery main fuse + holder (F-11)                         | bom/bom_estimated_items.csv row 125                                                                                                                                                                                                  |
| 12V battery disconnect (SW-12V-BATT)                                 | bom/bom_estimated_items.csv row 124                                                                                                                                                                                                  |
| 12V branch panel and blade fuses (F-10)                              | bom/bom_estimated_items.csv row 16                                                                                                                                                                                                   |
| Camper audio source/head-unit branch (AUDIO-HU / 12V-12)             | bom/bom_estimated_items.csv rows 189, 192, and 193                                                                                                                                                                                   |
| Camper audio powered-sub branch (AUDIO-SUB)                          | bom/bom_estimated_items.csv rows 191 and 192                                                                                                                                                                                         |
| SmartShunt external OEM fused red lead (OEM-SHUNT)                   | bom/bom_estimated_items.csv row 23 (included with SmartShunt kit)                                                                                                                                                                    |
| PV string fuses + holder (F-09A/B/C) and spares                      | bom/bom_estimated_items.csv row 106                                                                                                                                                                                                  |

## Assumptions and Open Items

- Wire sizing above assumes copper conductors and enclosed vehicle routing.
- F-09A/B/C and the old 3S3P fuse count are modeling placeholders only; final PV fusing waits until solar modules/stringing are selected after shore and alternator charging are working.
- Do not complete the owned Littelfuse 178.6150.0001 empty housings. Use the Mersen USCC1/ATDR15 branch for the PH-VAN red lead. The Wakespeed shunt is on the common battery-negative path after the SmartShunt, uses no 5A sense fuses, and must be configured Shunt at Battery .
- SW-12V-BATT remains manual-only in Phase 1; no automatic LVD behavior is assumed.
- Final lock for F-11 still requires explicit 12V buffer battery/BMS continuous discharge-current confirmation.
- F-15 exists to protect the smaller-gauge control wire between Ford Upfitter #3 and the WS500 brown ignition/enable wire; the factory 25A upfitter circuit protection is not the final wire-protection device for that branch.
- Orion F-05/F-06 lock: the active topology uses F-05 , one 40A MEGA body-marked at least 58VDC in Lynx Slot 4, feeding the existing 6 AWG Orion input directly. F-06 is retired. This deliberately prioritizes simple feeder protection over strict adherence to Victron's 20A table recommendation; do not stack an inline fuse after the Lynx fuse.
- Camper audio 12V load: KMC2 source branch and PTRTP10 powered-sub branch are downstream 12V loads, not new 48V branches. The PTRTP10 40A branch can exceed Orion 30A continuous output during heavy bass; rely on the 12V buffer battery for transients and validate sustained loud-use behavior before assuming all 12V loads can run at max simultaneously.
- Orion holder correction: no standalone input holder is required. The purchased 178.6150.0001 housing-only parts and 30A/58V MIDI stock remain unused; do not salvage or stack them into the final branch. Torque the Slot 4 MEGA hardware and Orion terminals to current manufacturer values, tug-test, and strain-relieve the existing 6 AWG pair.